

GEORGE JOBI PERANGATTU

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PROFESSIONAL SUMMARY

Robotics and autonomous systems graduate student with software, controls, sensor integration, and robot-systems experience relevant to electrical and cross-functional robotics engineering. Hands-on background with cameras, LiDAR, RGB-D sensors, actuators, electronics-adjacent integration, PID control, closed-loop control, kinematics, CAD-to-URDF workflows, ROS/Gazebo simulation, and real-hardware robot testing. Comfortable working across mechanical, electrical, hardware, and software boundaries to prototype, debug, test, document, deploy, and improve robotic systems in manufacturing and logistics contexts.

EDUCATION

M.S. Robotics and Autonomous Systems, AI Concentration, Arizona State University, Tempe, AZ - Graduating Jul 2026

B.S. Computer Science, Arizona State University, Tempe, AZ - Graduated May 2025

TECHNICAL SKILLS

Robotics Systems: Sensors, actuators, cameras, LiDAR, RGB-D cameras, sensor integration, kinematics, PID control, closed-loop control, robot navigation, ROS/ROS 2, Gazebo, CARLA, URDF

Hardware-Adjacent Engineering: CAD robot design, constrained joints, robotics data logging, simulation-to-real validation, control logic, manufacturing-process simulation, electronics integration context, sensors, actuators, testing, deployment, debugging, test planning, technical documentation

Software: Python, C++, C, Linux/Ubuntu, Git, CMake, Docker, CI/CD, GDB, testing, algorithms, data structures

Perception and ML: PyTorch, TensorFlow, computer vision, point cloud processing, object detection, segmentation, depth estimation, tracking, BEV perception

ROBOTICS PROJECTS

Spider CAD Robot: Multi-Legged Robot Design, ASU Robotics Coursework - Spring 2026

- Designed a multi-legged robot in CAD with constrained joints, validating kinematics and preparing a URDF-exportable model for simulation, controls, and hardware-software integration experiments.
- Considered actuator, motor, sensor, and joint constraints while preparing the model for ROS/Gazebo workflows.

Parrot MiniDrone: Autonomous Robot Navigation on Real Hardware, ASU Robotics Coursework - Spring 2026

- Developed real-hardware autonomous navigation using PID control and sensor fusion of inertial and visual data.
- Built automated test sequences, logged sensor data, and evaluated navigation performance under real-world constraints.
- Added safety systems for reliable autonomous operation and validated behavior in simulation and deployment settings.

Factory Automation System for Semiconductor Process Simulation, ASU Applied Project - Starting Jun 2026

- Planned an industrial automation system integrating robotic sensors, LiDAR, cameras, actuators, and control logic for manufacturing-process simulation.
- Focused on efficient robotics pipelines connecting sensing, control, and automated system behavior.

3D Motion Planning for Robot Navigation in Gazebo, ASU Robotics Coursework - Fall 2025

- Developed Python 3D RRT motion planning and integrated the planner with ROS/Gazebo for collision-free robot navigation tests.
- Ran evaluations across simulated environments and optimized trajectory quality through post-processing.

Efficient TransFuser: Multi-Task Sensor Fusion for Autonomous Navigation, ASU CSE 598 Research Project - Spring 2026

- Designed a camera-LiDAR fusion architecture for autonomous navigation, improving BEV segmentation by 25.4% and held-out loss by 8.7%.
- Worked with LiDAR, RGB, point clouds, depth estimation, and robotics perception pipelines relevant to intelligent robot feedback loops.

WORK EXPERIENCE

Backend Developer Intern, DigiClips Media, Tempe, AZ - Aug 2024 to May 2025

- Developed Python/PyTorch computer vision models and optimized data/training pipelines handling 10M+ daily queries.
- Reduced latency by 45% and improved detection accuracy by 30% through testing and performance-oriented debugging.

Full Stack Developer Intern, Odoo, Dubai, UAE - May 2025 to Aug 2025

- Refactored Python backend systems for production applications and collaborated with engineering teams.

Full Stack Developer Intern, TicketDex, Tempe, AZ - Jan 2025 to May 2025

- Delivered backend features and testing infrastructure, reducing latency by 30%.